in the gravitational field, so-called primordial gravitational waves. Inflationary cosmology and the cyclic model differ in this regard, one way in which there is a chance that they may be distinguished experimentally.

12. Quantum mechanics ensures that there is always a nonzero probability that a chance fluctuation will disrupt the cyclic process (e.g., one brane twists relative to the other), causing the model to grind to a halt. Even if the probability is minuscule, sooner or later it will surely come to pass, and hence the cycles cannot continue indefinitely.

## Chapter 14

1. A. Einstein, "Vierteljahrschrift für gerichtliche Medizin und öffentliches Sanitätswesen" 44 37 (1912). D. Brill and J. Cohen, *Phys. Rev.* vol. 143, no. 4, 1011 (1966); H. Pfister and K. Braun, Class. Quantum Grav. 2, 909 (1985).

2. In the four decades since the initial proposal of Schiff and Pugh, other tests of frame dragging have been undertaken. These experiments (carried out by, among others, Bruno Bertotti, Ignazio Ciufolini, and Peter Bender; and I. I. Shapiro, R. D. Reasenberg, J. F. Chandler, and R. W. Babcock) have studied the motion of the moon as well as satellites orbiting the earth, and found some evidence for frame dragging effects. One major advantage of Gravity Probe B is that it is the first fully contained experiment, one that is under complete control of the experimenters, and so should give the most precise and most direct evidence for frame dragging.

3. Although they are effective at giving a feel for Einstein's discovery, another limitation of the standard images of warped space is that they don't illustrate the warping of time. This is important because general relativity shows that for an ordinary object like the sun, as opposed to something extreme like a black hole, the warping of time (the closer you are to the sun, the slower your clocks will run) is far more pronounced than the warping of space. It's subtler to depict the warping of time graphically and it's harder to convey how warped time contributes to curved spatial trajectories such as the earth's elliptical orbit around the sun, and that's why Figure 3.10 (and just about every attempt to visualize